Operating Systems: CPU Scheduling & Process Management

Department: Computer Science & Engineering | **Semester:** 4 | **Subject Code:** CS402

Instructor: Department of Computer Engineering | **Source:** CampusVerse Study Bank

1. Introduction to CPU Scheduling

CPU scheduling is the process by which the operating system decides which of the processes in the ready queue is to be allocated the CPU processing core. The goal of CPU scheduling is to maximize CPU utilization, throughput, and minimize turnaround time, waiting time, and response time.

2. Comparative Analysis of Scheduling Algorithms

Algorithm	Preemptive?	Pros	Cons / Drawbacks
First-Come, First-Served (FCFS)	No	Simple to implement, fair order	Convoy Effect, high average wait time
Shortest Job First (SJF)	Optional	Optimal minimum average wait time	Starvation of long jobs, CPU burst prediction difficult
Round Robin (RR)	Yes	Fair CPU sharing, responsive for time-sharing	High context-switching overhead if quantum is too small
Priority Scheduling	Optional	Prioritizes mission-critical tasks	Indefinite blocking / starvation (resolved via Aging)

3. Solved Numerical Example: Round Robin (Time Quantum = 2ms)

Consider three processes: P1 (Burst Time = 5ms), P2 (Burst Time = 3ms), P3 (Burst Time = 1ms). All arrive at time 0.
Gantt Chart Execution: [0-2: P1] -> [2-4: P2] -> [4-5: P3] -> [5-7: P1] -> [7-8: P2] -> [8-9: P1]
Completion Times: P1 = 9ms, P2 = 8ms, P3 = 5ms.
Average Turnaround Time: (9 + 8 + 5) / 3 = 7.33 ms.
Average Waiting Time: [(9-5) + (8-3) + (5-1)] / 3 = (4 + 5 + 4) / 3 = 4.33 ms.

Exam Tips: Always draw the Gantt chart clearly with accurate timestamps before computing completion, turnaround, and waiting times.